

LITERATURE REVIEW

AIML Based De-Smoking / De-Hazing Algorithm

A study of deep-learning, CNN, attention, Transformer and task-aware approaches for image restoration under haze and smoke degradation

Prepared for: Project / Seminar Literature Review

Focus: Remote sensing, single-image dehazing, smoke-aware restoration and downstream vision

The review covers eight research papers, including three selected papers on remote sensing / haze-aware vision and five additional AIML-based dehazing papers.

Academic Year: 2026

1. Introduction

Haze and smoke are important sources of image degradation because suspended particles scatter and absorb light before it reaches the camera or sensor. The resulting images usually show lower contrast, colour distortion, reduced visibility and loss of fine structures. In remote sensing, the problem is more difficult because images contain small objects, large spatial areas and non-uniform atmospheric conditions. These effects can reduce the reliability of later tasks such as object detection, classification, segmentation and change analysis.

Traditional methods generally depend on the Atmospheric Scattering Model and manually designed priors such as the Dark Channel Prior. These approaches are useful because they provide a physical interpretation of haze formation, but their assumptions may not hold uniformly across complex scenes. AIML-based methods learn degradation and restoration patterns directly from data. CNNs provide strong local feature extraction, attention mechanisms improve feature selection, and Transformers provide better long-range interaction. More recent work also connects dehazing with the downstream task so that image restoration does not remove useful semantic information.

The selected literature therefore represents a progression from physically guided CNN models, to end-to-end and multi-scale networks, and finally to remote-sensing-specific Transformers and task-aware frameworks. The main objective of this review is to identify what has already been solved, what limitations remain, and where a practical AIML-based de-smoking/de-hazing system can contribute.

Review Objectives

- Study representative AIML methods for single-image haze/smoke removal.
- Compare physical-model, CNN, attention and Transformer approaches.
- Identify techniques useful for remote-sensing images and downstream vision tasks.
- Examine common limitations such as domain shift, detail loss, computation and real-world haze diversity.
- Formulate a research gap for a practical de-smoking/de-hazing system.

2. Literature Survey

1. Remote Sensing Single Image Dehazing With Histogram Degradation Representation Learning

Authors: Zhongmin Zhu, Tianyu Song, Yusi Huang, Weitao Han, Xinghui Xia, Ling Ma **Year:** 2026

Venue: IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

Main method: HRSformer: Histogram-guided Transformer; HDRB + MEMB

- Zhu et al. (2026) address haze in remote sensing images where smoke, thin clouds and atmospheric scattering reduce contrast and fine details.
- The proposed HRSformer uses a Histogram Degradation Representation Block (HDRB) to group spatially separated pixels with similar degradation patterns.
- A Multi-Expert Modulation Block (MEMB) adaptively selects spatial or channel experts to recover local texture at different scales.
- The method is designed to overcome the limited long-range modeling of CNNs and the restricted attention windows used by many earlier Transformers.
- Experiments on public remote-sensing dehazing benchmarks report stronger quantitative and visual performance than existing methods.

Key strength: Strong global degradation modeling and fine-detail reconstruction.

Observed limitation: Transformer-style global modeling can be computationally demanding for edge or real-time deployment.

2. HAE-Net: Haze-Aware Remote Sensing Object Detection With Local–Global Defogging

Authors: Hai Liao, Song Chen, Yan Liang, Hang Zhou, Fan Min **Year:** 2026

Venue: IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing

Main method: HAE-Net: Local and Global Defogging + Haze-Aware Detector + YOLO11s

- Liao et al. (2026) focus on remote-sensing object detection when haze causes loss of texture, edges, colour information and small-object cues.
- The paper identifies a practical problem with treating dehazing only as preprocessing: a visually clear image may still lose features needed by an object detector.
- Its Local and Global Defogging (LGD) network combines global context with local detail restoration to reduce feature loss.
- The Haze-Aware Detector uses multi-scale feature fusion and Normalized Wasserstein Distance loss so that dehazing remains useful for detection.
- The authors also introduce HAE-DIOR and report an mAP50 of 68.3% on that dataset, showing the value of task-aware dehazing.

Key strength: Connects image restoration directly with the downstream detection task.

Observed limitation: The framework is more complex than a standalone image-enhancement model and is tied to detection-oriented evaluation.

2. Literature Survey (continued)

3. Fire Segmentation in Complex Environments: Learning Representations for Reflection, Haze, and Day-Night Adaptation

Authors: Jong-Hyun Kim **Year:** 2025

Venue: IEEE Access

Main method: Single RGB pipeline: 3C correction + haze removal + reflection removal + U-Net + day/night adaptation

- Kim (2025) studies flame segmentation in real scenes where smoke/haze, reflections and changing illumination can confuse RGB-based vision systems.
- The work combines colour correction, haze removal and reflection suppression before learning the final flame segmentation with a lightweight U-Net.
- A Dark Channel Prior based step is used for smoke/haze removal, showing how a traditional prior can still be useful inside a learning-oriented pipeline.
- CycleGAN-based day-to-night adaptation is included to reduce errors caused by large illumination differences between daytime and nighttime images.
- The study reports an all-component IoU of 85.2% and emphasizes real-time deployment with a single RGB camera.

Key strength: Handles multiple visual disturbances rather than haze alone and keeps the input hardware simple.

Observed limitation: The application is fire segmentation, so the learned representation is not specifically optimized for remote-sensing dehazing.

4. DehazeNet: An End-to-End System for Single Image Haze Removal

Authors: Bolun Cai, Xiangmin Xu, Kui Jia, Chunmei Qing, Dacheng Tao **Year:** 2016

Venue: IEEE Transactions on Image Processing

Main method: CNN-based transmission-map estimation with Maxout and BReLU

- Cai et al. (2016) introduced DehazeNet as an early deep-learning system for single-image haze removal.
- Instead of directly predicting the clear image, the network estimates the medium transmission map from a hazy input.
- The estimated transmission is then used with the atmospheric scattering model to reconstruct a haze-free image.
- The architecture uses Maxout units for haze-related feature extraction and a bilateral rectified linear unit (BReLU) for improved restoration.
- The paper established an important bridge between hand-designed physical priors and trainable CNN representations.

Key strength: Simple and interpretable connection between CNN learning and the atmospheric scattering model.

Observed limitation: It depends on transmission estimation and the atmospheric model, which can be less flexible for complex non-uniform remote-sensing haze.

2. Literature Survey (continued)

5. AOD-Net: All-in-One Dehazing Network

Authors: Boyi Li, Xiulian Peng, Zhangyang Wang, Jizheng Xu, Dan Feng **Year:** 2017

Venue: IEEE International Conference on Computer Vision (ICCV)

Main method: Lightweight end-to-end CNN based on a reformulated atmospheric scattering model

- Li et al. (2017) proposed AOD-Net to simplify the dehazing pipeline by reformulating the atmospheric scattering model.
- The network directly learns a clean image rather than estimating transmission and atmospheric light as separate stages.
- Its lightweight CNN structure makes it easier to embed dehazing into larger computer-vision systems.
- The authors also demonstrated that coupling AOD-Net with Faster R-CNN can improve object detection under hazy conditions.
- This paper is relevant to AIML de-smoking because it shows how a compact learned restoration module can support downstream vision tasks.

Key strength: End-to-end and relatively lightweight; suitable as a baseline for practical systems.

Observed limitation: The model still uses a reformulated physical model and may struggle with spatially varying, complex haze.

6. Densely Connected Pyramid Dehazing Network

Authors: He Zhang, Vishal M. Patel **Year:** 2018

Venue: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

Main method: DCPDN: transmission estimation + atmospheric-light estimation + dehazing + joint discriminator

- Zhang and Patel (2018) proposed DCPDN as an end-to-end network that jointly learns important components of the atmospheric scattering process.
- The architecture contains a densely connected pyramid network for transmission estimation and a separate atmospheric-light estimation component.
- A dehazing stage reconstructs the clean image, while a joint discriminator encourages visually plausible restoration.
- Dense connections and multi-level pyramid pooling are used to preserve information across scales and improve edge-related details.
- The work demonstrates a hybrid direction in which physical constraints and adversarial learning are integrated in one restoration framework.

Key strength: Jointly learns physical quantities and image restoration with multi-scale feature flow.

Observed limitation: The multi-component architecture is heavier than lightweight CNN baselines and can be costly for resource-limited deployment.

2. Literature Survey (continued)

7. GridDehazeNet: Attention-Based Multi-Scale Network for Image Dehazing

Authors: Xiaohong Liu, Yongrui Ma, Zhihao Shi, Jun Chen **Year:** 2019

Venue: IEEE/CVF International Conference on Computer Vision (ICCV)

Main method: Trainable preprocessing + grid-based multi-scale backbone + post-processing

- Liu et al. (2019) proposed GridDehazeNet as an end-to-end CNN for single-image dehazing without relying on a fixed hand-crafted atmospheric prior.
- The trainable preprocessing module learns useful inputs, while the grid backbone performs multi-scale feature estimation.
- Attention mechanisms help the network adaptively combine information from different scales and focus on dehazing-relevant features.
- A post-processing module is added to reduce artifacts in the reconstructed image.
- The reported results show strong performance on both synthetic and real-world hazy images, making the model a useful benchmark for AIML dehazing.

Key strength: Strong multi-scale representation and attention-based feature fusion.

Observed limitation: The grid structure increases model complexity and does not by itself solve the synthetic-to-real domain gap.

8. GridDehazeNet+: An Enhanced Multi-Scale Network With Intra-Task Knowledge Transfer for Single Image Dehazing

Authors: Xiaohong Liu, Zhihao Shi, Zijun Wu, Jun Chen **Year:** 2022

Venue: IEEE Transactions on Intelligent Transportation Systems

Main method: Enhanced GridDehazeNet + spatial-channel attention + domain translation + intra-task knowledge transfer

- Liu et al. (2022) extended GridDehazeNet to address the domain-shift problem between synthetic training images and real hazy images.
- The enhanced network keeps the multi-scale grid design and adds spatial-channel attention for adaptive feature fusion.
- Synthetic data are translated toward the real-image distribution and then used for fine-tuning, reducing the mismatch between training and deployment conditions.
- An intra-task knowledge-transfer mechanism memorizes useful synthetic-domain knowledge and transfers it to the translated data.
- The study reports improved performance on several synthetic benchmarks and stronger real-world results after fine-tuning.

Key strength: Directly addresses generalization from synthetic haze to real haze.

Observed limitation: Domain translation and knowledge-transfer stages increase the training pipeline complexity.

3. Comparative Literature Survey Table

The table compares the papers according to the input/application, learning strategy, major contribution and the limitation most relevant to a practical AIML de-smoking/de-hazing system.

Paper	Year	Approach	Main contribution	Main limitation
Zhu et al. – HRSformer	2026	Transformer + histogram	Long-range degradation representation; multi-expert local detail recovery	Higher computational demand
Liao et al. – HAE-Net	2026	LGD + YOLO11s	Task-aware dehazing preserves detection features	More complex, detection-specific pipeline
Kim – Fire Segmentation	2025	RGB + preprocessing + U-Net	Joint haze, reflection and day/night handling	Application-specific to fire segmentation
Cai et al. – DehazeNet	2016	CNN + transmission map	Early learned transmission estimation	Physical-model dependence
Li et al. – AOD-Net	2017	Lightweight end-to-end CNN	Direct clean-image prediction; useful for downstream tasks	Limited flexibility for complex haze
Zhang & Patel – DCPDN	2018	Dense pyramid + GAN	Joint transmission, atmospheric light and restoration	Heavy multi-stage architecture
Liu et al. – GridDehazeNet	2019	Attention + multi-scale grid CNN	Strong scale fusion without fixed hand-crafted prior	Model complexity / domain gap
Liu et al. – GridDehazeNet+	2022	Grid + attention + knowledge transfer	Improves synthetic-to-real generalization	More complex training pipeline

4. Comparative Analysis

4.1 Evolution of the methods

The literature shows a clear shift from estimating physical haze variables toward learning features directly from images. DehazeNet learns transmission while still using the atmospheric model. AOD-Net simplifies this by reformulating the model and predicting the clear image in one network. DCPDN then jointly learns several physical components with dense multi-scale features. GridDehazeNet moves further toward data-driven restoration using attention and multi-scale feature fusion. GridDehazeNet+ adds domain adaptation and intra-task knowledge transfer, highlighting that generalization is as important as benchmark accuracy.

4.2 Importance of local and global information

CNNs are effective at recovering local textures and edges, but remote-sensing haze may have broad spatial variation. HRSformer explicitly models long-range relationships using histogram-guided degradation representation. HAE-Net also combines local and global defogging because object detection needs both scene context and small object details. This suggests that a useful practical system should not rely only on local convolutional features.

4.3 Generalization to real conditions

A repeated limitation across the literature is the gap between synthetic training haze and real haze. GridDehazeNet+ directly addresses this using distribution translation and knowledge transfer. Remote-sensing images add another challenge because haze is spatially non-uniform and can occur with thin clouds, smoke or illumination changes. Therefore, evaluation only on synthetic paired datasets is not enough for a robust field system.

4.4 Preservation of useful information

A dehazed image should not simply look brighter or clearer; it should preserve edges, textures, colours and object-level information. HAE-Net makes this issue explicit by optimizing dehazing for object detection. Kim's work similarly shows that haze and reflection removal must be combined with illumination adaptation in complex scenes. For an AIML de-smoking system, detail preservation should therefore be treated as a design requirement rather than only a visual-quality outcome.

4.5 Practical deployment

There is a trade-off between restoration quality and computational cost. Lightweight CNNs such as AOD-Net are attractive for embedded use, while Transformer and multi-stage models generally provide richer global representations at higher cost. A practical proposed system should therefore balance accuracy, parameter count, inference time and memory usage rather than selecting a model only from PSNR/SSIM.

5. Research Gap

Based on the reviewed papers, the following gaps are identified for AIML-based de-smoking/de-hazing, especially when the target application includes remote sensing or real-time visual analysis:

Gap 1 – Real-world smoke/haze diversity

Many deep models are trained using synthetic haze or controlled datasets. Real smoke can be spatially irregular, coloured, dense and mixed with illumination changes.

Gap 2 – Dehazing versus semantic preservation

A model can improve PSNR or visual clarity while still removing edges or small-object cues. Restoration and downstream task quality should be considered together.

Gap 3 – Local-global balance

CNNs capture local information well, whereas Transformers improve global interaction but can be computationally expensive. A balanced architecture is still desirable.

Gap 4 – Mixed degradation

Real scenes may contain haze/smoke together with reflection, low light, colour shift or thin cloud. A single-degradation model may not generalize well to such combinations.

Gap 5 – Efficient deployment

High-performing multi-stage and Transformer models may not be suitable for low-resource or edge devices. Lightweight feature extraction and efficient attention remain useful research directions.

Gap 6 – Evaluation beyond image quality

PSNR and SSIM do not fully describe whether the restored image is useful for detection, segmentation or classification. Task-aware metrics should be included where possible.

Possible research direction

A suitable proposed direction is a single-image AIML de-smoking/de-hazing framework that combines lightweight CNN feature extraction, attention-based local-global fusion and a detail-preserving reconstruction stage. Training can use a mixture of synthetic and real degraded images, while the loss can combine reconstruction quality with edge/detail preservation. If a downstream task is included, a task-aware loss can further reduce the risk of producing visually clear but semantically weak images.

6. Proposed System Concept

The proposed system is described at the literature-review stage, so no experimental performance values are claimed here. The idea is to use the strengths identified across the eight papers while keeping the architecture suitable for AIML implementation.

Suggested processing pipeline

Input degraded image → preprocessing/normalization → CNN shallow feature extraction → local-global attention module → multi-scale feature fusion → detail-preserving reconstruction → enhanced de-smoked/dehazed image → optional downstream detection/segmentation

Main design choices

1. **Single RGB input:** keeps the system practical for ordinary cameras and remote-sensing image inputs.
2. **Local feature branch:** preserves edges, texture and small structures using convolutional features.
3. **Global feature branch:** captures broad haze distribution using efficient attention or Transformer-style interaction.
4. **Multi-scale fusion:** combines coarse atmospheric information with fine object-level details.
5. **Residual reconstruction:** learns the degradation component and adds/restores it relative to the input.
6. **Detail-aware loss:** combines image reconstruction with edge/structural preservation.
7. **Optional task-aware objective:** if object detection or segmentation is the final goal, include a downstream loss.

7. Traditional vs. Proposed System – PPT Summary

The following points can be directly converted into presentation slides. The proposed system is a research direction derived from the literature and should not be presented as experimentally validated until testing is completed.

Aspect	Traditional / Existing System	Proposed AIML System
Input	Single hazy/smoky image; often assumes a specific degradation model.	Single RGB degraded image with support for mixed smoke/haze conditions.
Core method	Hand-crafted priors, atmospheric model, or separate estimation stages.	Learned CNN + attention/local-global feature fusion.
Feature extraction	Mostly manually designed image statistics.	Automatically learned multi-level spatial and contextual features.
Haze distribution	Often uses fixed assumptions about transmission/atmospheric light.	Learns spatially varying degradation from training data.
Detail preservation	Can introduce halos, colour distortion or oversmoothing.	Residual and detail-aware reconstruction is intended to preserve edges and textures.
Generalization	Can degrade when scene assumptions change.	Train with diverse synthetic + real degradations and augmentation/domain adaptation.
Downstream task	Often treated as a separate stage after enhancement.	Optional task-aware loss can jointly consider detection/segmentation usefulness.
Computation	Traditional priors may be fast but iterative optimization can be costly.	Target lightweight architecture and efficient attention for practical inference.

Suggested PPT slide order

Slide 1: Problem statement → Slide 2: Existing methods → Slide 3: Literature survey → Slide 4: Research gap → Slide 5: Proposed architecture → Slide 6: Training and evaluation → Slide 7: Expected advantages → Slide 8: Conclusion.

8. Summary

The literature indicates that AIML-based dehazing has moved from physical-prior estimation toward end-to-end feature learning, multi-scale attention and Transformer-based global representation. DehazeNet and AOD-Net established compact CNN baselines, while DCPDN introduced a stronger joint physical-learning framework. GridDehazeNet and GridDehazeNet+ improved multi-scale attention and real-world adaptation. The two recent remote-sensing papers, HRSformer and HAE-Net, show that remote-sensing dehazing increasingly needs global degradation understanding and task-aware feature preservation.

The fire-segmentation study adds another useful insight: real scenes may contain haze together with reflection and illumination changes, so a robust de-smoking pipeline should consider more than simple contrast enhancement. Overall, the strongest research opportunity is not merely to make an image look clearer, but to build a system that restores visibility, preserves important structures, generalizes to real degradation and remains computationally practical.

For a student AIML project, a reasonable next step is to implement a baseline such as AOD-Net or a small U-Net/CNN, establish PSNR/SSIM and visual comparisons, and then add an attention or multi-scale module. Real and synthetic degraded images should be tested separately so that generalization can be measured.

9. References

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Search and verification note

Bibliographic details and method summaries were checked against publicly available paper records, IEEE-related records, and publisher/open-access pages. The wording in this report is a concise academic synthesis rather than a reproduction of the source papers.